

USED CAR PRICE PREDICTOR

Machine Learning Web Application for
Predicting Used Car Prices

Project Overview

The Used Car Price Predictor is a machine learning web application that predicts the selling price of used cars based on multiple features. Built with Python, Flask, and Scikit-Learn, it demonstrates a real-world application of ML for price prediction.

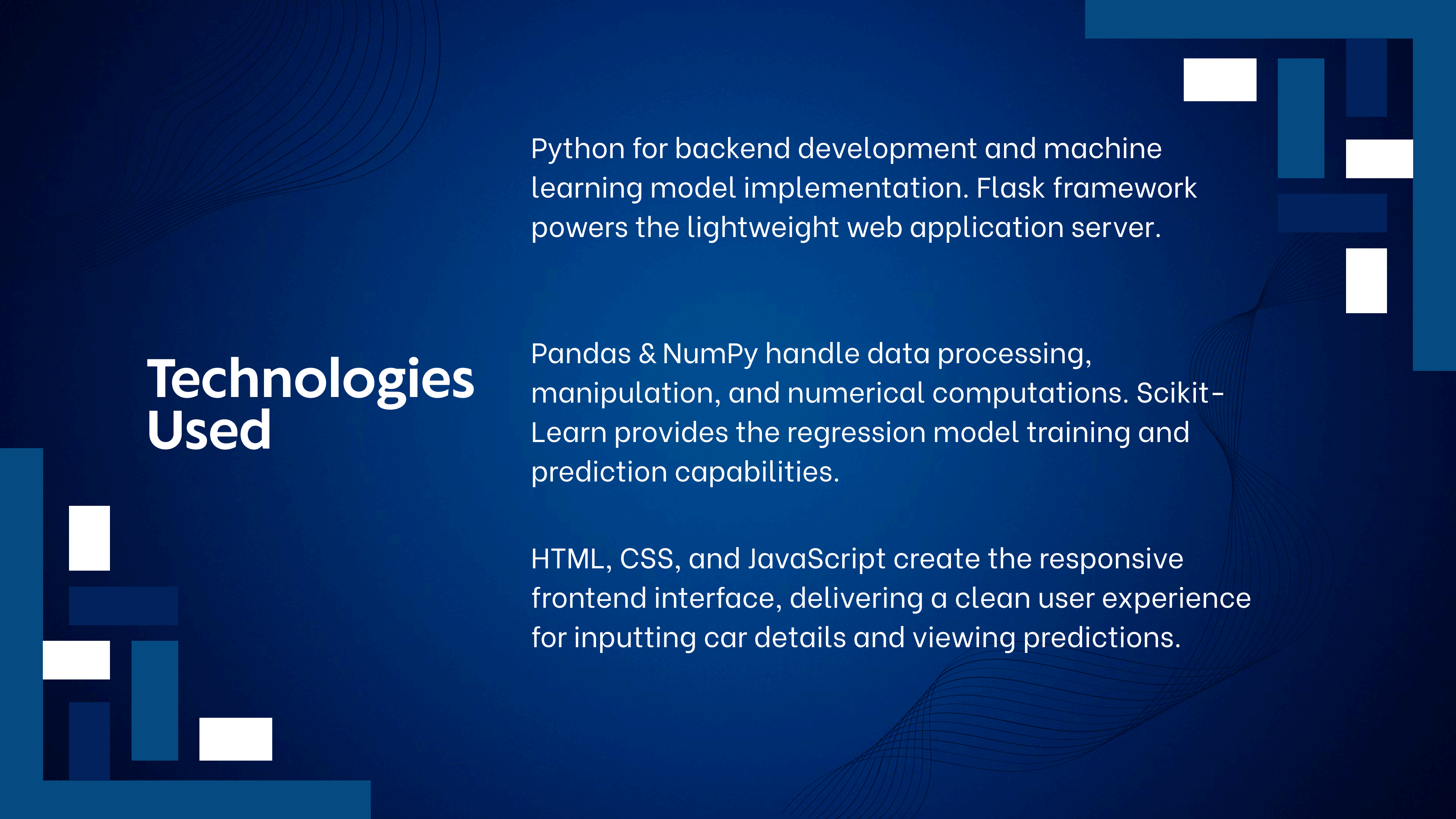
The system offers a simple and user-friendly web interface where users can input car details and instantly receive accurate price predictions. This project showcases how machine learning can solve practical problems in the automotive market.

Key Features

Predicts selling prices using a trained ML regression model powered by Scikit-Learn, providing accurate price estimates based on multiple car attributes.

Web interface built with Flask offers a simple, user-friendly experience for entering car details and receiving instant predictions.

Automatic model training runs if trained model is missing. Dataset included in repository ensures full reproducibility after cloning.



Technologies Used

Python for backend development and machine learning model implementation. Flask framework powers the lightweight web application server.

Pandas & NumPy handle data processing, manipulation, and numerical computations. Scikit-Learn provides the regression model training and prediction capabilities.

HTML, CSS, and JavaScript create the responsive frontend interface, delivering a clean user experience for inputting car details and viewing predictions.

Project Structure

1

DATA/
CARDEKHO_DATASET.CSV

2

STATIC/
STYLE.CSS, SCRIPT.JS

3

TEMPLATES/
INDEX.HTML

4

UTILS/
PREPROCESS.PY,
DOWNLOAD_DATASET.PY

5

APP.PY
TRAIN_MODEL.PY

6

REQUIREMENTS.TXT
README.MD

7

.GITIGNORE



Dataset Overview

- 1 CAR BRAND & MODEL
- 2 VEHICLE AGE
- 3 KILOMETERS DRIVEN
- 4 FUEL & TRANSMISSION

- 5 ENGINE & POWER
- 6 MILEAGE & SEATS
- 7 SELLING PRICE

How the System Works

1 USER INPUT

2 FLASK PROCESSING

3 AUTO-TRAIN
CHECK

4 PREDICTION

5 DISPLAY RESULT

Installation and Setup

1 CLONE
REPOSITORY

2 NAVIGATE TO FOLDER

3 INSTALL
DEPENDENCIES

4 RUN APPLICATION

5 OPEN BROWSER

Model Training Details

If the trained model file is not found, `train_model.py` runs automatically to generate it. The script follows a streamlined workflow to ensure the prediction system is always ready.

Script Workflow: Loads the CarDekho dataset → Preprocesses data using `utils/preprocess.py` → Trains a regression model with Scikit-Learn → Saves the trained model for prediction use.

Summary and Conclusion

The Used Car Price Predictor successfully combines machine learning and web technologies to deliver accurate price predictions for used vehicles. By leveraging Python, Flask, and Scikit-Learn, this project demonstrates the practical application of ML in solving real-world pricing challenges.

The user-friendly web interface makes complex predictions accessible to anyone, while the modular codebase ensures maintainability and scalability. This foundation opens doors for future enhancements, including expanded datasets and improved model accuracy.

The background is a deep blue gradient. In the top-left corner, there are several thin, white, concentric wavy lines. In the top-right corner, there is a cluster of white and light blue rectangles of various sizes. In the bottom-left corner, there is another cluster of white and light blue rectangles. In the bottom-right corner, there are more thin, white, concentric wavy lines. The text "THANK YOU" is centered in the middle of the image in a large, white, sans-serif font.

THANK
YOU